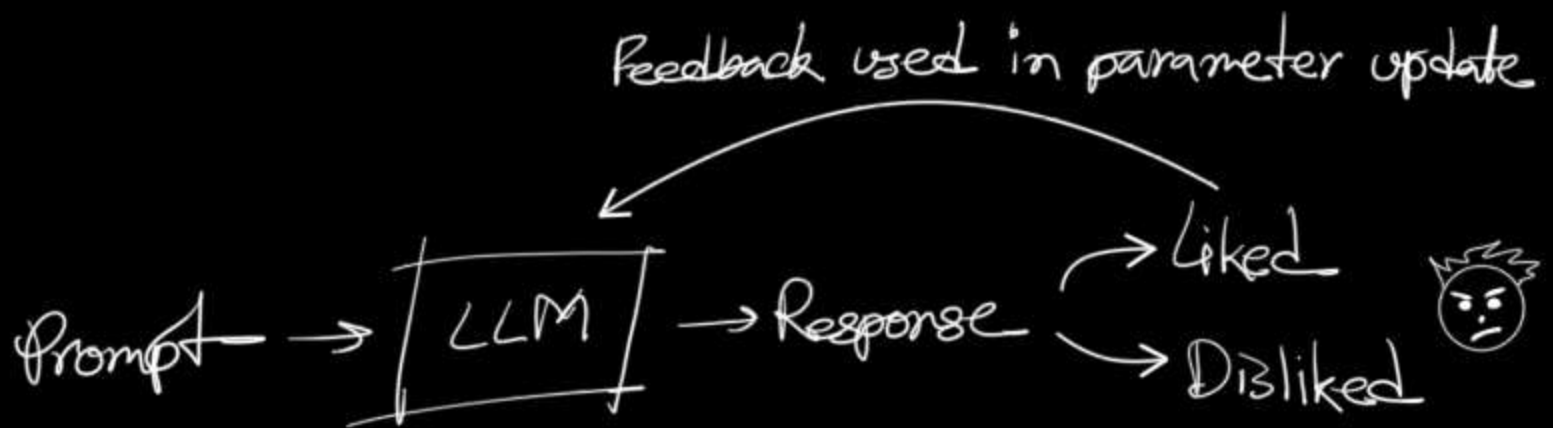


# RLHF

## Reinforcement Learning from Human Feedback

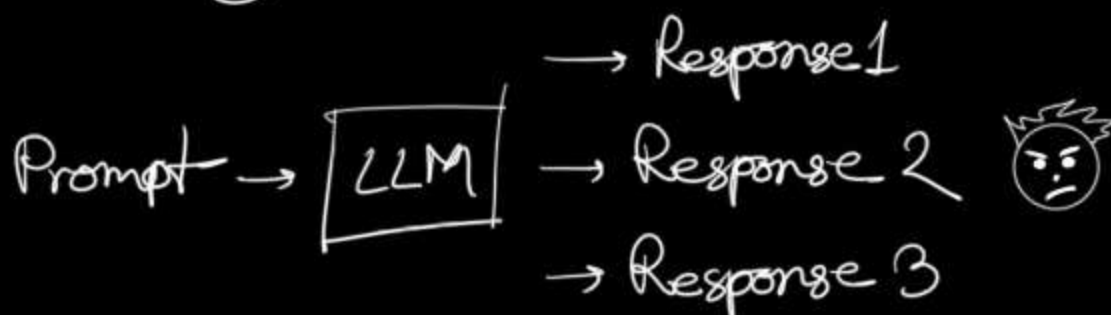
↳ Training LLMs on human preferences



But it is a slow process. So, OpenAI replaced human with a reward model, which was trained on human preferences.

## Reward model

↳ Proxy for human preferences



Giving rating is more complex than ranking. Based on some guidelines humans ranked the generated outputs from the same prompt.

Pair-wise preference dataset is created from the rankings & reward model is trained.

How to update the parameters?

Proximal Policy Optimization (PPO) → Will study later

One big limitation with RLHF:

The trained model will be only as good as reward model & the reward model is only as good as preference data.

# DPO

## Direct Policy Optimization

↳ Reformulating RLHF as a SL problem

How it works?

First we need a preference dataset →

| Prompt | Winner | Looser |
|--------|--------|--------|
|        |        |        |
|        |        |        |
|        |        |        |

Prompt → LLM → Probability of preferred response  
→ Probability of dispreferred response

So we can update the model's parameter to increase the probability of generating the preferred response & decrease the probability of generating dispreferred response.